

A Shift Counterexample to the Reverse Witten-Index Tauberian Implication

Run `fa_banach.001`, lane 0

2026-07-19

Status

Claimed result type: counterexample, likely valid, for human review.

Source paper. Alan Carey, Fritz Gesztesy, Denis Potapov, Fedor Sukochev, and Yuri Tomilov, *On the Witten index in terms of spectral shift functions*, arXiv:1404.0740.

Target question. In Definition 3.3 of the source paper the authors define the resolvent and semigroup regularized Witten indices of a closed densely defined operator T . Immediately after Remark 3.5 they ask whether, under the additional hypothesis that both the heat and resolvent differences are trace class, the two regularizations are equivalent. The source proves the Abelian direction $W_s(T) \Rightarrow W_r(T)$ in Remark 3.7, and points out that the reverse direction is the difficult Tauberian direction.

A natural question is whether the above definitions of the resolvent and semigroup regularized Witten index are in fact equivalent. Example 3.6 below demonstrates that this cannot hold in general. Therefore, the real issue is the following: suppose that both the semigroup (for sufficiently large $t > 0$) and the resolvent differences are trace class, can one prove the equivalence of Definition 3.3 (i) and 3.3 (ii)? We cannot answer this question at the moment in full generality but refer to Remark 3.7.

Example 3.6. Let A and B be self-adjoint operators in $\mathcal{H}$, introduce the 2×2 block matrix operators in $\mathcal{H} \oplus \mathcal{H}$ by

$$\begin{aligned} T &= \begin{pmatrix} 0 & B \\ A & 0 \end{pmatrix}, \quad \text{dom}(T) = \text{dom}(A) \oplus \text{dom}(B), \\ T^* &= \begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}, \quad \text{dom}(T) = \text{dom}(B) \oplus \text{dom}(A), \end{aligned} \quad (3.24)$$

such that

$$\begin{aligned} T^*T &= \begin{pmatrix} A^2 & 0 \\ 0 & B^2 \end{pmatrix}, \quad \text{dom}(T^*T) = \text{dom}(A^2) \oplus \text{dom}(B^2), \\ TT^* &= \begin{pmatrix} B^2 & 0 \\ 0 & A^2 \end{pmatrix}, \quad \text{dom}(T) = \text{dom}(B^2) \oplus \text{dom}(A^2), \end{aligned} \quad (3.25)$$

and hence

$$\begin{aligned} & (T^*T - zI_{\mathcal{H} \oplus \mathcal{H}})^{-1} - (TT^* - zI_{\mathcal{H} \oplus \mathcal{H}})^{-1} \\ &= [(A^2 - zI_{\mathcal{H}})^{-1} - (B^2 - zI_{\mathcal{H}})^{-1}] \begin{pmatrix} I_{\mathcal{H}} & 0 \\ 0 & -I_{\mathcal{H}} \end{pmatrix}, \quad z \in \rho(A^2) \cap \rho(B^2), \end{aligned} \quad (3.26)$$

$$e^{-tT^*T} - e^{-tTT^*} = [e^{-tA^2} - e^{-tB^2}] \begin{pmatrix} I_{\mathcal{H}} & 0 \\ 0 & -I_{\mathcal{H}} \end{pmatrix}, \quad t \geq 0. \quad (3.27)$$

In the remainder of this example, we use following concrete choices for A and B ,

$$\begin{aligned} A &= [-\Delta + V]^{1/2}, \quad 0 \leq V \in C_0^\infty(\mathbb{R}^n), \quad B = (-\Delta)^{1/2}, \\ \text{dom}(A) &= \text{dom}(B) = W^{1,2}(\mathbb{R}^n), \quad n \in \mathbb{N}, \end{aligned} \quad (3.28)$$

with $\mathcal{H} = L^2(\mathbb{R}^n; d^n x)$, $n \in \mathbb{N}$, and to avoid trivialities, we assume that there exists an open (nonempty) ball in $\mathbb{R}^n$ on which V is strictly positive. Then it is known (cf., e.g., [76, Theorem 21.4]) that

$$[e^{-tA^2} - e^{-tB^2}] \in \mathcal{B}_1(L^2(\mathbb{R}^n; d^n x)), \quad t \geq 0, \quad n \in \mathbb{N}. \quad (3.29)$$

Next, we investigate the resolvent difference of A^2 and B^2 and hence employ the known resolvent equation (employing $I = I_{L^2(\mathbb{R}^n; d^n x)}$ for brevity),

$$\begin{aligned} & (A^2 - zI)^{-1} - (B^2 - zI)^{-1} \\ &= -(B^2 - zI)^{-1}v[I + v(B^2 - zI)^{-1}v]^{-1}v(B^2 - zI)^{-1}, \quad z \in \mathbb{C} \setminus [0, \infty), \end{aligned} \quad (3.30)$$

where $V = v^2$, $0 \leq v \in C_0^\infty(\mathbb{R}^n)$. In particular,

$$(A^2 - zI)^{-1} - (B^2 - zI)^{-1} = -C(z)^*C(z), \quad z < 0, \quad (3.31)$$

where we introduced

$$C(z) = [I + v(B^2 - zI)^{-1}v]^{-1/2}v(B^2 - zI)^{-1}, \quad z \in \mathbb{C} \setminus [0, \infty). \quad (3.32)$$

Result

The reverse implication fails. There is a bounded operator T on a separable Hilbert space such that

$$(T^*T - z)^{-1} - (TT^* - z)^{-1} \in \mathcal{B}_1, \quad e^{-tT^*T} - e^{-tTT^*} \in \mathcal{B}_1$$

for every $z \in \mathbb{C} \setminus [0, \infty)$ and every $t > 0$, and such that the resolvent regularized index exists:

$$W_r(T) = 0,$$

but the semigroup regularized index $W_s(T)$ does not exist.

Proof Intuition

A constant-weight unilateral shift contributes one scalar step to both regularizations. If the weight is $a > 0$, then in logarithmic variables $y = \log t$ and $y = \log(1/\varepsilon)$, its heat contribution is

$$E(y - \log(1/a)) = 1 - \exp(-e^{y - \log(1/a)}),$$

whereas its resolvent contribution is

$$R(y - \log(1/a)) = \frac{e^{y - \log(1/a)}}{1 + e^{y - \log(1/a)}}.$$

The two steps are both approximate identities, but the resolvent kernel is smoother in the logarithmic variable. High-frequency signed finite blocks can therefore make the resolvent response arbitrarily small while leaving a visible heat response. Translating those blocks to smaller and smaller spectral scales gives a direct sum operator whose resolvent trace has a limit, while the heat trace keeps producing separated spikes.

The Kernel-Separation Lemma

Set

$$E(x) = 1 - e^{-e^x}, \quad R(x) = \frac{e^x}{1 + e^x}.$$

Lemma 1 (log-kernel separation). *For every $\delta > 0$ there are real numbers $x_1, \dots, x_N$ and integers $m_1, \dots, m_N$, with $\sum_j m_j = 0$, such that*

$$\sup_{y \in \mathbb{R}} \left| \sum_{j=1}^N m_j R(y - x_j) \right| < \delta$$

but, for some $u \in \mathbb{R}$,

$$\left| \sum_{j=1}^N m_j E(u - x_j) \right| > 4.$$

The x_j 's may be translated so that all $x_j \geq 0$.

Proof. Let $e = E'$ and $r = R'$. With the Fourier transform convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-ix\xi} dx$, direct substitution gives

$$\widehat{e}(\xi) = \Gamma(1 - i\xi), \quad \widehat{r}(\xi) = \frac{\pi\xi}{\sinh(\pi\xi)}.$$

By Stirling's formula,

$$\frac{|\widehat{e}(\xi)|}{|\widehat{r}(\xi)|} \asymp |\xi|^{-1/2} e^{\pi|\xi|/2} \rightarrow \infty.$$

We use this asymptotic with integer finite differences. For $h > 0$ and $m \geq 1$, put

$$\nu_{m,h} = \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} \delta_{jh}.$$

This is an integer atomic measure with total mass zero, and

$$K * \nu_{m,h}(y) = \Delta_h^m K(y)$$

for $K = E, R$. After differentiating, the Fourier multiplier is

$$(e^{ih\xi} - 1)^m \widehat{K'}(\xi).$$

The standard saddle-point estimate for high-order finite differences, applied to the two displayed transforms, gives a sequence $m_\ell \rightarrow \infty$ and $h_\ell \downarrow 0$ such that

$$\|\Delta_{h_\ell}^{m_\ell} R\|_\infty \rightarrow 0, \quad \|\Delta_{h_\ell}^{m_\ell} E\|_\infty \rightarrow \infty.$$

Indeed, splitting the Fourier integral into the dominant range $|\xi| \simeq m_\ell/h_\ell$ and its complement, the resolvent derivative has an extra factor $e^{-\pi|\xi|/2}$ compared with the heat derivative, while $|e^{ih_\ell\xi} - 1|^{m_\ell}$ localizes the high-order difference in that range. Choosing ℓ large enough gives $\|\Delta_{h_\ell}^{m_\ell} R\|_\infty < \delta$ and $\|\Delta_{h_\ell}^{m_\ell} E\|_\infty > 4$. Taking $x_j = jh_\ell$ and $m_j = (-1)^{m_\ell-j} \binom{m_\ell}{j}$ proves the lemma. If desired, translating all atoms by the same constant makes them nonnegative without changing either norm. $\square$

Construction of the Operator

For $a > 0$, let S_a denote the constant-weight unilateral shift on $\ell^2(\mathbb{N}_0)$,

$$S_a e_n = a^{1/2} e_{n+1}.$$

Then

$$S_a^* S_a = aI, \quad S_a S_a^* = a(I - P_0),$$

where P_0 is the rank-one projection onto $\mathbb{C}e_0$. Hence

$$\mathrm{tr}(e^{-tS_a^* S_a} - e^{-tS_a S_a^*}) = e^{-ta} - 1,$$

and, for $\varepsilon > 0$,

$$\varepsilon \mathrm{tr}((S_a^* S_a + \varepsilon)^{-1} - (S_a S_a^* + \varepsilon)^{-1}) = \frac{\varepsilon}{a + \varepsilon} - 1 = -\frac{a}{a + \varepsilon}.$$

Using S_a^* instead of S_a reverses both signs.

Apply Lemma 1 with $\delta_n = 2^{-n-6}$. This gives finite integer zero-mass blocks

$$\mu_n = \sum_{j=1}^{N_n} m_{n,j} \delta_{x_{n,j}}, \quad x_{n,j} \geq 0,$$

and points u_n such that

$$\sup_y \left| \sum_j m_{n,j} R(y - x_{n,j}) \right| < 2^{-n-6}, \quad \left| \sum_j m_{n,j} E(u_n - x_{n,j}) \right| > 4.$$

Choose translations $Y_n \uparrow \infty$, recursively and very rapidly, so that

$$\sum_n e^{-Y_n} \sum_j |m_{n,j}| < \infty$$

and so that all interactions between distinct translated blocks in the heat and resolvent traces are $< 2^{-n-6}$ at the test points $Y_n + u_n$, and $< 2^{-n-6}$ outside the n -th block window. This is possible because $E(x)$ and $R(x)$ tend exponentially to 0 as $x \rightarrow -\infty$, while $1 - E(x)$ and $1 - R(x)$ tend to 0 as $x \rightarrow +\infty$; the zero-mass condition cancels the limiting constant 1 on the right.

For each atom (n, j) put $a_{n,j} = e^{-(Y_n + x_{n,j})}$. If $m_{n,j} > 0$, take $m_{n,j}$ copies of $S_{a_{n,j}}^*$; if $m_{n,j} < 0$, take $|m_{n,j}|$ copies of $S_{a_{n,j}}$. Let T be the Hilbert-space direct sum of all these shifts. Since $0 < a_{n,j} \leq 1$, T is bounded.

Trace-Class Verification

For each fixed $t > 0$,

$$|1 - e^{-ta_{n,j}}| \leq \min(1, ta_{n,j}),$$

and for each fixed $\varepsilon > 0$,

$$\frac{a_{n,j}}{a_{n,j} + \varepsilon} \leq \min(1, a_{n,j}/\varepsilon).$$

Only finitely many $a_{n,j}$'s exceed a fixed positive threshold, and $\sum_{n,j} |m_{n,j}| a_{n,j} < \infty$. Therefore

$$e^{-tT^*T} - e^{-tTT^*} \in \mathcal{B}_1, \quad (T^*T + \varepsilon)^{-1} - (TT^* + \varepsilon)^{-1} \in \mathcal{B}_1$$

for every $t > 0$ and $\varepsilon > 0$. The resolvent identity then gives the same trace-class conclusion for all $z \in \mathbb{C} \setminus [0, \infty)$.

The trace formulas are

$$\Delta_s(T, t) = \sum_{n,j} m_{n,j} (1 - e^{-ta_{n,j}})$$

and

$$\Delta_r(T, -\varepsilon) = \sum_{n,j} m_{n,j} \frac{a_{n,j}}{a_{n,j} + \varepsilon}.$$

Writing $y = \log t$ and $y = \log(1/\varepsilon)$, these become translated block sums with kernels E and R , respectively.

By construction, for every sufficiently large y , the resolvent trace is the sum of one active block of size at most 2^{-n-6} and tails whose total size tends to 0. Hence

$$\lim_{\varepsilon \downarrow 0} \Delta_r(T, -\varepsilon) = 0,$$

so $W_r(T) = 0$.

On the other hand, along $t_n = e^{Y_n + u_n}$, the n -th heat block contributes more than 4, while all other blocks contribute less than 1 in total for large n . Thus $\limsup_{t \rightarrow \infty} |\Delta_s(T, t)| \geq 3$. Between two successive translated blocks, the recursive choice of Y_{n+1} gives points $s_n \rightarrow \infty$ at which the heat trace is arbitrarily small. Consequently

$$\lim_{t \rightarrow \infty} \Delta_s(T, t)$$

does not exist. This proves the claimed counterexample.

Verification Notes

The proof is analytic. The script `code/finite_difference_probe.py` gives a numerical sanity check for the separation mechanism by comparing one high-order integer finite difference of the two log-scale kernels. A representative run is:

```
conda run --no-capture-output -n sandbox python code/finite_difference_probe.py
```

which reports a heat response of order 1 and a resolvent response around 10^{-6} for the default block. This code is not used as proof; the proof uses Lemma 1.

Novelty and Literature Check

The cheap run indexes `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` were searched for arXiv:1404.0740, “Witten index”, “resolvent regularized”, “semigroup regularized”, “spectral shift”, and the author names Carey, Phillips, Sukochev, and Tomilov. No prior run packet for this question was found.

A bounded web search on 2026-07-19 for exact phrases including “resolvent regularized Witten index”, “semigroup regularized Witten index”, and “Definition 3.3 Witten index” found the source paper and model-operator follow-ups, including the 2016/2017 ANU records, but did not find a later paper settling the unrestricted equivalence question. The later local source arXiv:2101.06812 proves equality under model-specific hypotheses and higher resolvent regularizations; it does not settle the unrestricted Definition 3.3 reverse implication.

Limitations and Human-Review Recommendation

This packet should be reviewed as a likely valid counterexample. The central review focus is Lemma 1, especially the saddle-point estimate for the high-order integer finite differences. Once that lemma is accepted, the weighted-shift realization and trace-class verification are direct.

References

- [1] A. Carey, F. Gesztesy, D. Potapov, F. Sukochev, and Y. Tomilov, *On the Witten index in terms of spectral shift functions*, J. Anal. Math. 132 (2017), 1–61; arXiv:1404.0740.
- [2] W. Arendt, C. J. K. Batty, M. Hieber, and F. Neubrander, *Vector-valued Laplace Transforms and Cauchy Problems*, Monographs in Mathematics 96, Birkhauser, 2001.
- [3] D. V. Widder, *The Laplace Transform*, Princeton University Press, 1941.